

Nominal Rigidities and Stabilization: Calvo Pricing → New Keynesian Model → Monetary Policy

A coherent, exam-style summary with consistent notation.

Calvo Pricing

Purpose. Provide a dynamic microfoundation for sticky prices: only a random subset of firms can reset prices each period.

Nominal rigidity (Calvo assumption).

- In each period, a fraction $\alpha \in (0, 1]$ of firms can reset prices; the rest keep their previous price.
- If a firm resets at t , it remains stuck with that price for k additional periods with probability $(1 - \alpha)^k$.

Key objects.

- Inflation: $\pi_t \equiv \log P_t - \log P_{t-1}$.
- Output gap: $\hat{x}_t \equiv \hat{Y}_t - \hat{Y}_t^n$, where $\hat{Y}_t^n$ is the (log-linearized) flexible-price/natural output.

Core implication: New Keynesian Phillips Curve (output-gap form).

$$\pi_t = \kappa \hat{x}_t + \beta \mathbb{E}_t[\pi_{t+1}], \quad \kappa \equiv \frac{\alpha [1 - (1 - \alpha)\beta]}{1 - \alpha} [\theta + \gamma - 1],$$

where $\beta \in (0, 1)$ is the discount factor and (θ, γ) summarize preferences/technology in the log-linearization.

Interpretation. Inflation is forward-looking (via $\mathbb{E}_t[\pi_{t+1}]$) and increases with economic slack (via $\hat{x}_t$), with slope κ pinned down by price stickiness α and structural parameters.

New Keynesian Model

Endogenous variables: output gap $\hat{x}_t$, inflation π_t , nominal interest rate i_t .

Exogenous drivers: natural interest rate r_t^n and cost-push shock μ_t .

Three-equation system.

$$\hat{x}_t = \mathbb{E}_t[\hat{x}_{t+1}] - \frac{1}{\theta} (i_t - \mathbb{E}_t[\pi_{t+1}] - r_t^n), \quad (\text{NK-IS})$$

$$\pi_t = \kappa \hat{x}_t + \beta \mathbb{E}_t[\pi_{t+1}] + \mu_t, \quad (\text{NKPC})$$

$$i_t = \phi_\pi \pi_t + \phi_x \hat{x}_t. \quad (\text{Taylor rule})$$

Determinacy (Taylor principle). Because $(\hat{x}_t, \pi_t)$ are forward-looking “jump” variables, a unique non-explosive equilibrium requires a sufficiently aggressive interest-rate response to inflation:

$$\phi_\pi + \frac{1 - \beta}{\kappa} \phi_x > 1 \quad (\text{and if } \beta \approx 1, \phi_\pi > 1 \text{ is the familiar benchmark}).$$

Interpretation.

- *Demand side (NK-IS):* the real interest rate gap (relative to r_t^n) moves the output gap.
- *Supply side (NKPC):* inflation depends on slack and expected future inflation; μ_t shifts the inflation–slack tradeoff.
- *Closure (Taylor rule):* policy feedback pins down expectations and rules out sunspots when the Taylor principle holds.

Monetary Policy

Policy objective (quadratic stabilization). A standard formulation is to choose sequences $\{\pi_t, \hat{x}_t, i_t\}_{t \geq 0}$ to minimize

$$\min_{\{\pi_t, \hat{x}_t, i_t\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t (\pi_t^2 + \lambda \hat{x}_t^2), \quad \lambda > 0,$$

subject to the NK-IS and NKPC constraints (and using i_t to satisfy NK-IS given $(\pi_t, \hat{x}_t)$).

Benchmark without cost-push shocks. If $\mu_t \equiv 0$, the first-best stabilizes both targets:

$$\pi_t = 0, \quad \hat{x}_t = 0, \quad i_t = r_t^n.$$

With cost-push shocks: commitment vs discretion. When $\mu_t \neq 0$, full stabilization of both π_t and $\hat{x}_t$ is generally impossible. The key distinction is how expectations are managed:

- **Commitment (history dependence):**

$$\hat{x}_t - \hat{x}_{t-1} = -\frac{\kappa}{\lambda} \pi_t.$$

- **Discretion (re-optimizing each period):**

$$\hat{x}_t = -\frac{\kappa}{\lambda} \pi_t.$$

Why rules are used (credibility + equilibrium selection).

- *Time inconsistency:* even if commitment is optimal, a central bank typically has incentives to deviate ex post, undermining credibility.
- *Implementability and determinacy:* interest-rate rules that satisfy the Taylor principle deliver a unique equilibrium and rely only on observable $(\pi_t, \hat{x}_t)$.

Coherence map

Calvo Pricing $\Rightarrow$ NKPC and (NKPC + NK-IS + Policy Rule) $\Rightarrow$ New Keynesian Model

New Keynesian Model $\Rightarrow$ Monetary Policy: rules vs. (infeasible) full commitment.

One-page comparison table

Model	Friction / control	Core equation(s)	Takeaway
Calvo Pricing	Random price adjustment: fraction α resets	$\pi_t = \kappa \hat{x}_t + \beta \mathbb{E}_t[\pi_{t+1}]$	Microfounds sticky prices; delivers forward-looking inflation dynamics
New Keynesian Model	Sticky prices + interest-rate rule	NK-IS, NKPC, Taylor rule	A complete system for $(\hat{x}_t, \pi_t, i_t)$; determinacy requires the Taylor principle
Monetary Policy	Stabilization tradeoff under shocks	$\min \mathbb{E}_0 \sum \beta^t (\pi_t^2 + \lambda \hat{x}_t^2)$ s.t. NK constraints	Commitment creates history dependence; discretion raises credibility/multiplicity issues; rules are practical